

Mid-Semester Exam Study Guide

BITS Pilani WILP · Second Semester 2025–2026 · All 4 Courses

Exam Dates at a Glance

Course Code	Course Name	Date	Session
AIMLC ZC416	Mathematical Foundations for ML	20 Jun 2026	Afternoon
AIMLC ZC418	Introduction to Statistical Methods	20 Jun 2026	Forenoon
AIMLC ZG565	Machine Learning	21 Jun 2026	Forenoon
AIMLC ZG511	Deep Neural Networks	21 Jun 2026	Afternoon

All mid-semester exams are **Closed Book** — no reference material permitted.

1. Mathematical Foundations for Machine Learning (AIMLC ZC416)

Mid-sem: 20 Jun 2026 (Afternoon) · Closed Book · Sessions 1–8

Sess.	Topic	Key Sub-topics
1	Linear Systems	Systems of equations, matrices, solving $Ax=b$
2	Vector Spaces	Linear independence, basis, rank
3	Analytic Geometry	Norms, inner products, distances, angles, orthogonality, orthonormal basis
4	Matrix Decomposition I	Determinant, trace, eigenvalues, eigenvectors, Cholesky decomposition
5	Matrix Decomposition II	Eigen decomposition, diagonalization, SVD, matrix approximation
6	Vector Calculus I	Univariate differentiation, partial derivatives, gradients of vector-valued functions
7	Vector Calculus II	Gradients of matrices, gradient identities, backpropagation, automatic differentiation
8	Vector Calculus III	Higher-order derivatives, multivariate Taylor series, maxima/minima for unconstrained optimization

2. Introduction to Statistical Methods (AIMLC ZC418)

Mid-sem: 20 Jun 2026 (Forenoon) · Closed Book · Sessions 1–8

Sess.	Topic	Key Sub-topics
1	Basic Probability & Stats	Central tendency, variability, 5-point summary, intro to probability
2	Basic Probability	Axioms of probability, mutually exclusive & independent events
3	Conditional Probability	Conditional probability, independent events, total probability
4	Bayes Theorem	Bayes theorem (with proof), Naive Bayes concept & problems
5	Probability Distributions I	Discrete & continuous RVs, expectation, mean, variance, joint distributions, transformation of RVs

Sess.	Topic	Key Sub-topics
6	Probability Distributions II	Bernoulli, Binomial, Poisson, Normal, t, F, Chi-square distributions
7	Hypothesis Testing	Random & stratified sampling, Central Limit Theorem, interval estimation, confidence levels
8	Review	Review of sessions 1–7

3. Machine Learning (AIMLC ZG565)

Mid-sem: 21 Jun 2026 (Forenoon) · Closed Book · Sessions 1–8

Sess.	Topic	Key Sub-topics
1	Introduction to ML	What is ML, taxonomy of learning, design a learning system, challenges
2	ML Workflow + Math Prelims	Linear algebra, calculus, probability, data preprocessing, model training & evaluation metrics
3	Linear Regression	Direct solution, gradient descent (batch/SGD/mini-batch), bias-variance decomposition
4	Linear Classification	Discriminant functions, decision theory, probabilistic discriminative classifiers, intro to logistic regression
5	Logistic Regression	Log-loss function, gradient descent, multi-class classification
6	Decision Trees	Information theory, entropy, ID3 construction, overfitting, MDL, continuous/missing attributes
7	Instance-Based Learning	k-NN, locally weighted regression (LWR), radial basis functions
8	Review	Review of sessions 1–7

4. Deep Neural Networks — UPDATED (AIMLC ZG511)

Mid-sem: 21 Jun 2026 (Afternoon) · Closed Book · Sessions 1–8 · Prof. Seetha Parameswaran (v2, Sep 2025)

Sess.	Topic	Key Sub-topics
1	Introduction & Motivating Example	Deep learning – what, why, where; wake word detection; key components (data, models, objective fn, optimization); supervised learning; interacting with environment
2	Artificial Neuron & Perceptron	Biological vs artificial neuron; connectionism model; perceptron; NOT/AND/OR gates; learning algorithm; linearly separable data; XOR problem (perceptron failure); code from scratch
3	Linear NNs for Regression	Single neuron regression; linear model with squared loss; mini-batch SGD training; prediction/inference; Auto MPG (UCI) code from scratch
4	Linear NNs for Classification	Single neuron (sigmoid + binary cross-entropy); multi-class (softmax + cross-entropy); mini-batch SGD; Breast Cancer Wisconsin (binary) & Iris (multi-class) code from scratch
5	Deep Feedforward Networks (MLP)	MLP solving XOR; hidden layers & non-linearity; forward propagation (vectorized); loss functions; backward propagation (vectorized); weight updates; impact of depth and width; code from scratch

Sess.	Topic	Key Sub-topics
6	Convolutional Neural Networks	Image data; invariance, locality, channels; convolution; learning a kernel; feature maps & receptive field; padding & stride; 1x1 convolution; pooling; multiple channels; LeNet end-to-end; code
7	Deep CNN Architectures	AlexNet (representation learning); VGGNet (VGG blocks); Network in Network (NiN); GoogLeNet (Inception blocks); ResNet (residual blocks)
8	Transfer Learning & Review	Transfer learning concepts; fine-tuning for transfer learning; review of sessions 1–7